ABSTRACT

This project concerns the design and fabrication of a manually operated sheet metal bending machine using a hydraulic bottle jack, pressing tool, dies, springs, and a supporting frame made from mild-steel channels and angles. The objective is to provide a simple, low-cost arrangement for bending sheet metal, with the original project identifying a target capacity of up to 4 mm sheet thickness. During operation, hydraulic pressure moves the piston and pressing tool against the workpiece supported on the table and die. The table height can be adjusted using lock nuts. After bending, the pressure-relief valve is released to return the piston. The report describes the components, working principle, design approach, cost-estimation method, assembly, applications, conclusions, and future scope. The source project lists an estimated material and other expenditure total of ₹15,147; this figure should be checked against original bills and quotations before formal use.

KEYWORDS

Sheet metal bending, hydraulic jack, punch, die, mild-steel frame, fabrication, cost estimation.

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CHAPTER 1 – INTRODUCTION

Sheet metal bending is a forming operation used to produce shapes such as flanges, channels, corrugations, and other formed parts. During bending, the sheet is subjected to bending stresses: material on the outside of the bend experiences tensile stress, while material on the inside experiences compressive stress. The bend radius describes the inside radius of curvature. In practical plastic bending, the neutral axis shifts toward the inside of the bend and local thickness reduction may occur.

1.1 Objectives of the Project

- To develop a bending machine intended to bend sheet metal up to 4 mm thick.
- To use a simple working principle.
- To reduce the time required for the bending operation.
- To keep the construction cost low.

1.2 Design Details

The design process requires consideration of alternative mechanisms, component choices, materials, forces, manufacturing methods, safety, and cost. The machine is intended as a straightforward manually operated arrangement for sheet-metal bending.

1.3 Features

- Low initial and tooling cost.
- Easy and quick setting.
- Versatility for different bending tasks.
- Repeatable bending when the workpiece and setup are positioned consistently.
- User-friendly operation and easy maintenance.
- Use of standard parts that are generally easier to source.

CHAPTER 2 – COMPONENTS

The source project identifies the following principal components: sheet metal, hydraulic jack, springs, dies, pressing tool / press-brake tooling, and mild-steel channel and angle sections.

2.1 Sheet Metal

Sheet metal is metal processed into thin, flat pieces that can be cut and formed. Common materials include steel, aluminium, copper, brass, nickel, and titanium. The suitable material and thickness depend on the intended part and the bending requirements.

2.2 Hydraulic Jack

A hydraulic jack uses pressurised, substantially incompressible fluid to generate force. Pumping the jack handle moves oil through check valves into the cylinder, raising the piston. Releasing the pressure-relief valve allows pressure to reduce and the piston to return. In this project, the hydraulic bottle jack provides the force that drives the pressing tool.

2.3 Tension Spring

A tension spring stores energy when stretched and returns toward its original length when the applied force is removed. Springs are included among the machine components; the source report does not specify their dimensions or spring rate.

2.4 Dies

Dies support and shape the workpiece during bending. The source report discusses several die types, including 90-degree, acute-angle, gooseneck, offset, hemming, seaming, radius, beading, curling, channel-forming, U-bend, box-forming, corrugating, multiple-bend, and rocker-type dies. The selected die geometry affects the bend profile and workpiece clearance.

2.5 Pressing Tool / Press Brake

A press brake forms a bend by pressing a workpiece between matching punch and die tooling. In this project, the pressing tool is fixed to the hydraulic piston-rod end. Industrial press brakes may use mechanical, pneumatic, hydraulic, or servo-electric force systems; the project uses a hydraulic jack arrangement.

2.6 Mild-Steel Channel and Angle

Mild-steel channel and angle sections are used to form the supporting structure and frame. The frame must support the bending load and keep the table, jack, and tooling aligned. The source report does not provide complete frame dimensions or strength calculations.

CHAPTER 3 – METHODOLOGY AND WORKING PRINCIPLE

3.1 Methodology

The machine is arranged around a hydraulic bottle jack and pressing tool. The principal tasks are selecting components, preparing the support frame and table, fitting the jack and pressing tool, positioning the dies, assembling the parts, and checking operation. Detailed fabrication dimensions and inspection records were not supplied in the source report.

3.2 Working Principle

- Place the hydraulic bottle jack on the base plate.
- Fix the pressing tool to the end of the hydraulic piston rod.
- Place the work table on the supporting frame and adjust its height using the lock nuts on either side.
- Close the pressure-relief valve and operate the pump rod. Hydraulic oil pressure raises the piston and moves the pressing tool against the workpiece.
- Once the required bend is achieved, release the pressure-relief valve so the piston can return to its original position.
- Remove the workpiece only after the tool has returned and the load is safely released.

3.3 Advantages and Limitations

Advantages stated in the source report include simple construction, easy maintenance, and low cost. A stated limitation is that continuous manual pumping can cause operator fatigue. Actual capacity, repeatability, and safe operating limits should be established through design checks and testing.

3.4 Applications

The machine is intended for sheet-metal work and fabrication tasks in small-scale industries and workshops. The source report also mentions cable-wire-making industries.

CHAPTER 4 – USES OF THE MACHINE

- Sheet-metal bending in small-scale industries.
- Fabrication work.

- Portable or straightforward workshop bending tasks, subject to the machine's verified capacity.
- Potential reduction in bending-operation cost where the machine is suitable for the workpiece.

CHAPTER 5 – DESIGN

5.1 General Requirements of Machine Design

- High productivity.
- Required shape and dimensional accuracy with suitable surface finish.
- Simple design.
- Safe and convenient controls.
- Low cost.
- Good appearance.

5.2 Design Procedure

The source report groups design work into functional design, product design, and engineering design. These stages define the required function, overall product arrangement, and detailed engineering choices needed for manufacture.

5.3 Design Considerations

- Clearly define the problem and the purpose of the machine.
- Select a mechanism that provides the required motion.
- Determine the forces and energy transmitted through machine elements.
- Select suitable materials for each element.
- Determine allowable stresses and sizes to avoid excessive deformation or failure.
- Consider productivity, demand, safety, and intended applications.
- Refine parts to support practical manufacture and assembly.
- Prepare assembly and detail drawings with material, dimensional, accuracy, and surface-finish specifications.

5.4 Drawings

The original report includes drawing sections labelled “Spring” and “Hydraulic Jack.” Insert or retain the approved component drawings and a complete assembly drawing here before final submission. The source file does not provide a complete dimensioned machine drawing in readable text.

CHAPTER 6 – COST ESTIMATION

Cost estimation forecasts the expenses involved in manufacturing a product. The source report describes material, machining, and labour costs as the principal cost categories.

6.1 Purpose of Cost Estimation

- Estimate a selling price for a quotation or contract.
- Check supplier quotations.
- Compare economical materials and manufacturing processes.
- Establish cost standards for production control.

6.2 Material Cost

Material costs include raw material for fabricated components and standard purchased parts. A bill of materials should list each component, dimensions, material, quantity, unit rate, and extended cost.

6.3 Machining and Labour Cost

Machining cost is estimated from the time required for each operation, including setup, inspection, and centring allowances, multiplied by the applicable machine-hour rate. Labour cost may be direct or indirect. The source report gives the general relationship:

TOTAL COST = MATERIAL COST + MACHINING COST + LABOUR COST

SELLING COST = MANUFACTURING COST + 10% PROFIT

6.4 Cost Summary

The source report lists the following expenditure. The total below is the arithmetic sum of the listed amounts and should be verified against actual purchase records.

Material / Item	Cost (₹)
Sheet metal	1,500
Hydraulic jack	5,500
Springs	600
Iron channel	2,500
Punch	3,677
Iron angle (dies)	300
Other cost	1,070
TOTAL	15,147

CHAPTER 7 – ASSEMBLY OF PARTS

The source report identifies the jack, springs, dies, C-channel, and overall machine structure as assembly items. A final assembly record should include each component's dimensions, material, quantity, joining method, and inspection status.

Component	Details available in source
Hydraulic jack	Used as the force-generating unit; detailed dimensions not specified.
Springs	Included; dimensions and spring rate not specified.
Dies / punch	Used to form the bend; detailed tooling dimensions not specified.
C-channel / angle sections	Used for the support structure; complete section sizes not specified.
Overall machine	A dimensioned assembly drawing and verified overall dimensions are not supplied in readable text.

CHAPTER 8 – COST EXPENDITURE

The listed component expenditure totals ₹15,147 based on the amounts in the source report. This is a reported estimate, not a verified current quotation or complete manufacturing cost; machining and labour costs may not be fully represented in the item list.

CHAPTER 9 – CONCLUSION AND FUTURE SCOPE

9.1 Conclusion

The project report describes the design and assembly of a hydraulic-jack-operated sheet metal bending machine. The operating arrangement is simple: the jack raises the pressing tool to bend the workpiece against the die, and the pressure-relief valve allows the piston to return after bending. The project provided practical experience in planning, purchasing, assembly, machining, and teamwork. The source report states that the machine was tried at the college and operated satisfactorily for its intended purpose.

9.2 Future Scope

Possible future work identified in the source report includes improving efficiency and adding features for commercial use. Before wider use, the design could be developed further with complete dimensioned drawings, load and frame-strength calculations, defined material and thickness limits, safer guarding and controls, a more ergonomic pumping arrangement, and documented repeatability and safety testing. These are recommendations for future development, not features confirmed in the source project.

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